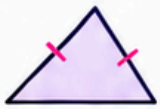
Geometry Triangles & Congruence Cheat Sheet

Unit Focus: *triangle facts, congruence, and proofs*

1 CLASSIFYING TRIANGLES

By Sides (lengths of sides)

Scalene
No congruent sides (all sides different).



Isosceles
Two congruent sides (exactly two sides the same length).



Equilateral
Three congruent sides (all sides the same length).

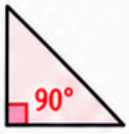


By Angles (measures of angles)

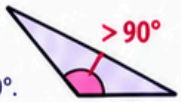
Acute
All three angles are less than 90°.



Right
One angle is exactly 90°.



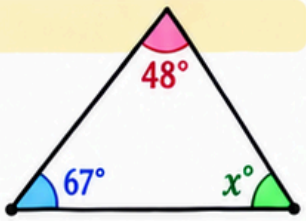
Obtuse
One angle is greater than 90°.



2 TRIANGLE SUM THEOREM

The measures of the three interior angles of a triangle always sum to 180°.

Example: Find the value of x.



Step 1: Set up an equation using the triangle sum.

$$48° + 67° + x = 180°$$

Step 2: Solve for x.

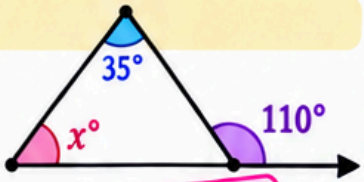
$$115° + x = 180°$$
$$x = 65°$$

Answer: $x = 65°$

3 EXTERIOR ANGLE THEOREM

An **exterior angle** of a triangle equals the sum of the **two remote interior angles** (the two *nonadjacent angles*).

Example: Find the value of x.



Step 1: Set up an equation using the exterior angle theorem.

$$110° = 35° + x$$

Step 2: Solve for x.

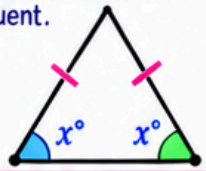
$$x = 75°$$

Answer: $x = 75°$

4 ISOSCELES TRIANGLE THEOREM

Isosceles Triangle Theorem

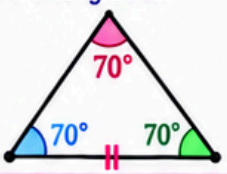
If two sides of a triangle are congruent, then the base angles (opposite those sides) are congruent.



Base angles are congruent!

Converse

If two angles of a triangle are congruent, then the opposite sides are congruent.



Opposite sides are congruent!

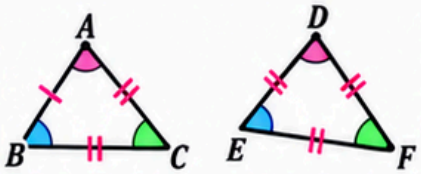
5 CONGRUENT FIGURES

Congruent figures have the same size and shape. Corresponding parts of congruent figures are congruent (corresponding sides and angles have the same measures).

Matching Order Matters!

If $\triangle ABC \cong \triangle DEF$, then:

- $A \leftrightarrow D$
- $B \leftrightarrow E$
- $C \leftrightarrow F$



$\triangle ABC \cong \triangle DEF$
means $\overline{AB} \cong \overline{DE}$, $\overline{BC} \cong \overline{EF}$, $\overline{AC} \cong \overline{DF}$
and $\angle A \cong \angle D$, $\angle B \cong \angle E$, $\angle C \cong \angle F$.

6 TRIANGLE INEQUALITY

The sum of ANY two side lengths in a triangle must be GREATER than the third.

Example: lengths 4, 6, and 8:

4 + 6 > 8; 4 + 8 > 6; 6 + 8 > 4.

So these lengths form a triangle.

But 2, 3, 6 cannot form a triangle:

2 + 3 is not greater than 6.

7 TRIANGLE SIDE RELATIONSHIPS

The longest side is opposite the largest angle. The shortest side is opposite the smallest angle. Equal sides have equal opposite angles.

Example: if angles A = 40°, B = 60°, and C = 80°, then AB is the longest side (opposite angle C).

8 QUICK REMINDERS & COMMON MISTAKES

DO REMEMBER:

- Use the triangle sum (180°) first when finding missing angles.
- Mark congruent sides and angles using tick marks and angle arcs.
- Keep track of corresponding order when naming congruent triangles.
- Use theorems (like isosceles, exterior angle, midsegment, etc.) when possible.
- A diagram can help, but your reasoning and statements must be correct.

COMMON MISTAKES:

- Forgetting to use the triangle sum before solving for an angle.
- Matching the wrong corresponding vertices in congruence (order matters!).
- Assuming a picture is drawn to scale (unless it is given).
- Confusing similar figures with congruent figures.
- Not justifying steps in a proof or missing key reasons.

Geometry Triangles & Congruence Cheat Sheet

More Practice: triangle congruence shortcuts and CPCTC

1 CONGRUENCE SHORTCUTS

SSS: three corresponding sides.

SAS: two sides and INCLUDED angle.

ASA: two angles and INCLUDED side.

AAS: two angles and a non-included side.

HL: right triangles with equal hypotenuses and one equal leg.

Corresponding vertices must match.

2 NOT ENOUGH INFORMATION

AAA proves similarity, not congruence.

SSA is ambiguous and does not generally prove congruence.

For SAS, the known angle must lie between the known sides.

For HL, both must be right triangles.

Do not use a diagram alone as proof.

3 WORKED SSS / SAS EXAMPLES

SSS Example:

AB = DE = 6, BC = EF = 8,

AC = DF = 10.

Therefore ABC and DEF are congruent by SSS, with A ↔ D, B ↔ E, C ↔ F.

SAS Example:

AB = DE, AC = DF,

and angle A = angle D.

Therefore ABC and DEF are congruent by SAS.

4 CPCTC EXAMPLE

CPCTC: Corresponding Parts of Congruent Triangles are Congruent.

First establish triangle congruence.

If triangle ABC is congruent to triangle DEF, then:

AB = DE, BC = EF, AC = DF

angle A = angle D

angle B = angle E

angle C = angle F

5 STRATEGIES & COMMON MISTAKES

✓ Mark equal sides with matching tick marks; mark equal angles with matching arcs.

✓ Write triangle names in corresponding order: ABC ↔ DEF.

✓ Use CPCTC only AFTER establishing congruence.

✗ AAA is not a congruence test.

✗ SSA is not generally a congruence test.

✗ Do not call an angle INCLUDED unless it is between the stated sides.